

DEDICATION

This Capstone project is dedicated to all the unwavering support, motivation, and love from all of our Families, Peers, Professors and IT Experts who helped our cause. For it is our achievement alone as we would not be able to thrive this far without you. To God almighty, your presence and guidance has given us hope and courage. For your grace, we have been blessed in our journey as we culminate our identity as an IT.

May this be an exhibit and fruition of all of our hard work and perseverance, the emotions that we shared and invested, the experiences we were able to accumulate and the memories we have built throughout the development. For whichever path we take, this will be a testimony of the precious time we shared.

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ETugma: Tutor Matching System with Assessment for Scribbles and Schemes Early Childhood Center

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Overview

Tutoring is a rising market in the education industry. It is the act of a teacher teaching a student outside of the class hours. However, tutors can find it hard to find students while students also have a hard time looking for a tutor that is suitable for them. This system provides a platform where students can reach out to tutors that can fit their preferences. The system uses Collaborative Filtering in which it sorts and filters the tutors based on the specifications provided by the student or client. It also provides a chat and email feature which lets both client and tutor communicate with each other. Overall, this system is an accessible and effective way for students to learn more through tutoring.

Categories and Subject Descriptors

CCS -> Information systems -> World Wide Web -> Web searching and information discovery -> Collaborative filtering.

General Terms

Management, Verification, Accessibility, Algorithm

Keywords

Matchmaking, Educational Web Application, Filtering

1. THE PROBLEM AND ITS BACKGROUND

1.1 Project Context

1.1.1 Introduction

“Education is the passport to the future, for tomorrow belongs to those who prepare for it today.” This is a quote from Malcolm X, an American human rights activist. Education is a basic need for every person, it brings them a

weapon in order to face different ordeals that they may face in their life. Schools are the main place where people get educated, it teaches them various lessons that can help them in different aspects of life. It upholds a class-style of learning in which multiple people are being taught by different teachers of varying subjects. However, an additional way of teaching can be used which is through tutoring. It is defined to be as teaching in a specific academic subject that usually takes place outside of school premises.

Tutoring is a special way of learning since many people are not accustomed to this method of learning. Its effectiveness is a beneficial way of learning extra material for a subject, based on a study done on Education of Roma Tre University¹⁹, tutoring has been proven to be effective in combating a student’s learning inefficiency. Tutoring as a whole can be treated as an advanced learning since it exceeds the current curriculum that is being taught by a school. It can help people who are struggling with a certain topic and people who want to learn more.

Nowadays in which education is truly needed due to the various information that are floating around everywhere, it may be from the internet or spread through spoken means. Being educated acts as a shield to protect oneself in being misled by false information that can lead them into believing the wrong thing. On a global scale, false news is prominent with 51 percent of fake news being seen and 44 percent in printed media¹.

Whether it be through classes or being tutored, having knowledge is still a power that each person should have in order to have a brighter tomorrow.

1.1.2 Background of the Study

In the field of academics, students often encounter a significant challenge when they are chasing for a better education: the hurdle of finding a suitable tutors. The same applies to tutors looking for a job, they often encounter a problem where they are having a hard time looking for clients or students because they would rely only on word of mouth or referrals. According to (Pockovic A. 2023), one disadvantage was mentioned that a referral employee's personality has a possibility of not fitting to the team. They may have skills but could be rigid and only want to implement methods from their previous employer². Having a limited option can be a problem to both users. Exclusively relying only on referrals can limit your choices and opportunities of finding a proper tutor and client. If the tutor might be far from your location, it might hinder your studies especially if onsite class is needed, which is why the overall compatibility of both students and tutors is necessary.

Despite the importance of having a tutor and their compatibility, there are only limited internet platforms available in the Philippines. The limited platforms available on the internet hinders students who are looking for additional classes. Having no proper system, students and tutors often lead to referral methods or in traditional methods which can cause inefficiencies and disadvantages. With the current educational system, it is more demanding as it was before. In the study of Laurie Kimbre (2019), principals mostly rely on traditional interviews when hiring teachers. But the traditional method has been known to be problematic because of the cons it gives such as validity and its reliability.

Scribbles and Schemes Early Childhood Center is a Tutoring center located in Biñan, Laguna and Silang, Cavite. It is established to educate children from preschool to Junior high school to nurture their academic skills and teach them to become independent learners. There are two ways for tutoring in the center, parents can enroll their children directly to the center or students can be tutored in a session. The subjects that are being offered by the center are the usual subjects that are being learned by preschool to highschool students such as Science, Math, and English.

The current situation of Scribbles and Schemes Early Childhood Center is falling behind in the online tutoring market as they do not have any platform, they could use except their Facebook page to promote or integrate their tutoring method online. The center also implements a manual way of recording by using a logbook as their way of collecting their data which is very unsafe and unreliable as the traditional method can pose many problems such as duplication of the data, Time consuming, security of the data, difficulty of retrieving data, and data entry errors. By using (ETugma), Scribbles and Schemes will have an online platform accessible to parents and tutor users as an advertising platform and learning platform. Logbooks will not be necessary anymore as there will be a database that can store and collect the data automatically.

1.2 Purpose and Description

This system's purpose is to give a satisfactory experience in learning for the tutor and the client. It provides that experience by doing the following:

- Provide a place for students to find tutors that can match their preferences.
- Help expand the tutoring market.
- Make transactions between tutor and student to be more efficient.
- Create a learning environment outside the class hours provided by the school.

1.3 Objectives of the Study

1.3.1 General Objective

The main objective of this study is to develop an accurate and reliable online tutor matching system that will aid the students to find a suitable tutor for them, likewise for tutors who are looking for students to become their clients.

1.3.2 Specific Objective

In order to meet the general objective, the following actions are done:

- To develop a system that will provide a list of prospective tutors allows users to search by expertise, location, fees, and etc.
- To verify the qualifications of the tutors that are registered in the system by submitting a softcopy of their document through the website and a hardcopy through the admin.
- To provide communication between clients and tutors through chat and direct messaging platforms.
- To spread the benefits of tutoring in the education of students

1.4 Scope and Limitation

This system will be built using the following languages and frameworks:

Front-end:

- HTML
- CSS
- JavaScript

Back-end:

- PHP
- JavaScript

Frameworks:

- Bootstrap
- Laravel

This system has five user levels which are the guests, tutors, parents, admin, and head admin. The actions they can do are as follows:

Tutors:

- Can adjust their tutoring schedules based on their availability.
- Notified if a student/client is booking them for a session.
- Can accept the applications of students/clients.
- Communicate with students/clients using chat or direct message.
- Can check the current status for each session that they have.
- Can send documents that can verify if their credentials are valid or not.
- An online payment is provided using GCash.
- Can assess the work done by the student.
- The progress of the student in each session will be evaluated by the tutor.
- Can be able to see the transactions done in the system for transparency.
- Can upload modules or assessments for their students to answer.

Parents/Clients:

- Find tutors that are based on their preferences which is sorted through collaborative filtering.
- Can communicate with tutors using chat or direct message.
- Can book tutors.
- Can be notified if tutors approve their booking.
- Can rate the services of the tutors and add reviews for the tutor.
- Can pay tutors through GCash.
- Students can submit their works or their scores in the system.
- Able to monitor the progress of the tutoring using a tracker in the system.
- Able to see the transaction done in the system for the transparency.
- Can register multiple children/students.

Admins:

- Can check the documents of an applying tutor to verify its validity.
- Can be able to view the transaction history of both students and tutors.

Head admin.

- Can view the transaction of both students and tutors.
- Can manage the users. If the user is inactive for about a year, violated a rule, or received multiple complaints/bad reviews, the account will be deactivated.
- Can check the ratings and reviews given by the clients to the tutors.

- Can update the contents of the system using the Content Management System (CMS)

Guests

- Can only view the homepage of the system.
- Can register as a tutor or client.
- Can view the about us of the system.

and lastly, to test the developed system using ISO 25010 Quality Model for web applications.

The limitations of the system will be the following:

- Subjects offered are only those provided by the Scribble and Schemes Early Childhood Center Binan Branch.
- Admins cannot change or manage the schedules agreed upon by both tutors and students.
- Tracking of the actual tutoring session is not done.
- This system will not have a mobile application.

1.5 Definition of Terms

Admins - the people who will manage the operations of the system.

Client - People who use the services of the system.

Collaborative Filtering - The algorithm that will be used for the matching system of the website. It uses the client's preferences to sort the choices.

Credentials - The document that the tutors need to submit to be a verified tutor.

CSS - A front-end language that will be used to design the website.

Education - The service that the website serves. It is the product of a tutor teaching a tutee.

GCash - An online payment app that will be used to accommodate the online payment module in the system.

HTML - A front-end language that will be used to design the website.

Javascript - A language that is commonly used for the front-end and back-end of a system.

Language - The set of instructions or notations made by the developers to create a code that can run the system.

Laravel - A web framework that is used to display the system.

Matching system - Algorithmic process designed to connect students looking for academic assistance with suitable tutors based on specific preferences.

PHP - A back-end language that is used for the functions of the system and modules.

Preferences - The specific instructions that can benefit and guide the clients in choosing a tutor.

Referrals - The spread of tutoring is done by telling other people about the tutor themselves.

Students - The people learning under the tutor.

Subject expertise - The topic that the tutor excels at and the topic the student wants to learn.

Tutoring - A way of learning outside the curriculum provided by the school.

Tutors - Individual with educational background seeking for clients/students.

Word-of-mouth- A referral way of recommending a tutor to students.

2. REVIEW OF RELATED LITERATURE AND STUDY/SYSTEMS

2.1 Local Literature

2.1.1 Philippine online tutoring platforms see surge in demand amid virus

During the pandemic, the demand for online tutoring rose due to schools closing. In countries in Asia, particularly China and Japan, tutors who provide English instruction are in high demand. The e-learning firms have seen growth with their shares and clients doubling during the pandemic. Companies like 51Talk have planned to expand on Filipino tutors in hopes to meet the demand and capitalize the market¹.

ETugma will be a way for tutors to keep up with current demand as it provides a platform for them to find clients in an efficient way.

2.1.2 Why More and More Filipinos Are Searching for Online Jobs

Participating in an online job has its benefits such as:

- It allows you to work from home which means commuting is avoided and transportation cost are reduced.
- It is more flexible in order to fit one's schedule.
- It pays more than conventional office-based roles.

With these benefits, more and more Filipinos began to search for more online jobs thus shrinking the market for traditional jobs².

The rise of online jobs here in the country leaves more opportunities for tutors to look for jobs in an online setting, making the learning experience become more comfortable between the tutors and students.

2.1.3 Here's How Fintech Is Modernising Digital Payments in the Philippines

Digital payments are continuously growing due to the Bangko Sentral ng Pilipinas and the government's push for financial inclusion. The creation of the National Retail

Payment System (NPRS) in 2015 has facilitated electronic payments here in the country. Companies like GCash, PayMaya, and Coins.ph are offering diverse digital payment options which helps the spread of mobile phones throughout the country³.

The system uses an online payment module using GCash as the main platform for online transactions. The popularity and easy usability of GCash makes it the main option in implementing an online payment module for ETugma.

2.1.4 The easy guide to employee background checks in the Philippines

Here in the Philippines, a background check is important when applying for a job. This helps verify a person's qualifications and reduces the risk of bringing in a person who can disrupt the operations of a company. Records like identity, reference check, work authorization, education history, etc. are checked by HR to see if the person applying has clean qualifications⁴.

Since freelance tutors can register to be a tutor on the website, a thorough background check should be done in order to keep the integrity of the tutors and the website itself. Trustworthy tutors are always the priority of the system as its goal is to provide a true learning experience for both tutors and students in ETugma.

2.1.5 Online Tutoring: An Emerging Market in the Philippines

Online tutoring is gaining more popularity in the Philippines due to many reasons. It offers a convenient and flexible way of learning environment for tutors and students since both people can access the comfort of their homes when interacting with each other. The specific subjects that can be learned are also an important factor in why people are leaning toward using online tutoring. Factors like breaking geographical barriers and cost-effectiveness also contribute to the rise of online tutoring in today's market⁵.

ETugma's matching system helps tutors to find more clients in the vast market nowadays. Since the usual method of finding clients is through referrals or word-of-mouth, it can be inefficient and time consuming for tutors to find students. This system will help them search in a more efficient way.

2.1.6 5 Reasons Why You Should Use Online Job Search Platforms

This article states that searching for jobs online can be more effective than finding it in a traditional way. It says that applicants can search with ease as jobs are posted online regularly and it can be filtered to match the applicant's choice. It also has benefits such as efficient employer lookout, sending out multiple applications at a single time, and communication would be easier⁶.

Teachers that are applying at ETugma can easily find their clients on the matching system provided by ETugma. This can help them find clients in a comfortable yet efficient way making their lives much easier.

2.1.7 MATATAG CURRICULUM

The MATATAG Curriculum is the new curriculum approved by the Department of Education or DepEd, it is implemented for the school year of 2024 to 2025 for K-10.

Since ETugma's main focus is the education of the Scribbles and Schemes Early Childhood Center's students, ETugma system will use this curriculum made by DepEd as a guide to provide an accurate and informative lessons for the respective student of the tutoring center³².

2.2 Foreign Literature

2.2.1 The importance of identity verification in the education sector

Identity verification is crucial in education to prevent fraud, comply with regulations like FERPA, protect students' privacy, and curb cheating. It enhances overall institutional security amid increasing cyber threats and improves efficiency by streamlining processes. Educational institutions must prioritize identity verification to safeguard sensitive information, maintain reputation, and ensure a secure learning environment.

By verifying the identity of the tutor who will apply to the website, we can prevent any fraudulence, protect the safety of students and to ensure the quality of teaching. The verification process involves checking the expertise of the tutors and credentials of tutors stating their background in education³.

Having a verified identity is crucial for the system of ETugma as it makes sure that the integrity of the website is secured, and reliable. Having bogus accounts in the website will make the transactions and process slow down due to different bogus accounts being recommended to students and vice versa where bogus students are booking for an appointment with a tutor.

2.2.2 Collaborative Filtering in Recommender Systems: Learn All You Need to Know

Collaborative filtering is a recommendation system that creates a prediction based on a user's previous behaviors. This type of algorithm is also known as a product-based collaborative shift. The main idea behind this approach is suggesting new items based on the closeness in the behavior of similar customers¹⁰.

The algorithm that will be used for ETugma is Collaborative Filtering in which students will be recommended a tutor based on the preferences that they have on what kind of tutor they want.

2.2.3 Exploring System Design: Instant Messaging Application

This article looks at the design decisions made when creating a messaging app, focusing on the special features that make these apps unique. Designing a real-time chat application requires careful consideration of several key requirements such as Real Time Communication, Message Persistence, and more¹¹.

A chat feature will be implemented in the web-based system of ETugma for smooth and efficient communication between the tutor and the student. In this chat feature, they can talk about whether they will conduct classes through F2F or online, the subjects that they will learn, the fees and locations, etc.

2.2.4 Online communication and customer review

Good online communication is crucial for businesses online. It includes talking to customers on social media, through emails, blogs, and live chats. Businesses need to have clear rules, check different channels regularly, reply quickly, train their staff, ask customers for reviews, manage those reviews well, and use positive reviews in promotions. This helps build strong relationships, a good reputation, and attracts more customers¹².

Admins of ETugma will keep track of the reviews that are garnered by the tutors in every session that is finished. This is to regularly check if the tutor is doing his/her job in a satisfactory way to the student. Good reputations for the tutor will be established due to trust that students have with his/her work as a tutor.

2.2.5 Effects of Virtual Tutoring on Young Readers

In this research, students who have received OnYourMark Education virtual tutoring have shown improvements in early literacy over students that have received usual instructions. The research showed that kindergarteners could sound 3+ more letter sounds than students who did not receive the tutoring¹³.

The benefits that students can get with online tutoring is helpful for them in school as it can give them an advanced knowledge about the subjects they learn in the curriculum. Tutoring counts as an extracurricular study for students and it can give a boost for learning.

2.2.6 Teacher Hiring: The disconnect between research based best practice and processes used by school principals

This research highlights that principals commonly use traditional interviews to hire teachers, despite their known limitations in predictability, validity, and reliability. Additionally, many principals lack consistent teacher selection processes and may change their criteria based on subjective opinions of importance, further reducing the likelihood of hiring the most successful teachers.

This proves why ETugma is efficient when finding tutor for students compared to the traditional method. By using a tutor matching process, this eliminates the time-consuming nature of the traditional method.

2.2.7 A Tutor's Guide to Student Assessment Online.

In this guide, it shows the different ways a teacher can assess his/her students in an online setting. Benchmark assessment, formative assessment, and summative assessments are the different methods shown in the guide. These types of assessments will help a teacher to see the progress of a student in class.

ETugma's system will integrate various assessments methods to cater to different learning styles and objectives. With an assessment module, ETugma can help teachers to

accommodate the works of their students and assess if they properly learned from that session²⁵.

2.3 Local Study/Systems

2.3.1 GCash payment API

In this system, it shows how to integrate GCash as a payment method for the system. It uses a checkout URL where customers need to authorize an amount, wait for the authorization to finish, and to create a payment resource in order to receive the authorized amount¹⁴.

GCash is the main platform for the online payment module of ETugma's system. Using an API can make the integration for the online payment module to be seamless for the transaction of the tutors and the clients.

2.3.2 GrabTutor: A web and mobile application for tutor appointment system with GPS security feature

This study aims to create a safe and efficient experience for students undergoing tutoring with a tutor appointment system and location tracking feature. A web and mobile application is used that is connected through a cloud server. A unique barcode is generated for each tutorial appointment and a barcode scanner that is embedded in the mobile app is given to the tutor. Once the tutorial starts by scanning the barcode provided to the tutor, the system will trigger the timer and a GPS coordinate to track the location. The results show that students are matched with the right tutors and the safety of students is secured¹⁵.

With the use of GrabTutor, the project is now clear that it will be benefiting the students and also tutors as instead of using the traditional way of locating tutor, there will now be a platform that can automatically look for your preferred tutor.

2.3.3 Designing and Implementing e-School Systems: An Information Systems Approach to School Management of a Community College in Northern Mindanao, Philippines

This system is about the implementation of a digital management system for the school. It keeps the records of each student's information, and it also processes and retrieves data in the system. This was made to reduce the workload of the school due to the large paperwork involved in storing the student's information¹⁶.

ETugma also uses a management system in which the admins can view the information of student and client such as the current sessions being done, student's reference, etc. while head admins can view and edit the information in the system.

2.3.4 The Effectiveness of Peer Tutoring in Enhancing Reading Comprehension of Ninth Grade Students

This study conducts experimental research that uses a pretest-posttest design to learn more about the topic. The data collected was from sixty Grade 9 students using a researcher-made questionnaire that had been verified by a panel of experts. The results showed that the students achieved the required level of reading comprehension. There was a significant difference in the reading comprehension of the students before and after the tutoring¹⁷.

This study shows that tutoring is an efficient way for the education of students as there are significant results in the knowledge of students when being tutored.

2.3.5 Integrating Randomized Suggestions into Collaborative Filtering Applied to Movie Recommendation

This study uses Collaborative Filtering to predict the user's preferences which includes the historical data and past interactions to recommend items they might like. Problems that can be encountered in Collaborative Filtering are cold-starts, lack of diversity, and popularity bias which is why a Linear Congruent Generator (LCG) is introduced. The issue of a cold start was resolved by using LCG and generating a set of random users and their favorite movies and the lack of diversity and popularity bias was resolved by sorting the movies from most rated to least rated. LCG was used throughout the start to finish of the study¹⁸.

ETugma's matching system will also use Collaborative Filtering in recommending tutors for students. It will also display the tutors that have frequent students and also the highest rated tutors in the system when a new student joins with no preferences inputted. This will guide them on the tutors that can be the most reliable to book.

2.4 Foreign Study/Systems

2.4.1 Enhance the quality of the university tutoring services to counteract undergraduate students' educational failure: a focus on the Department of Education of Roma Tre University.

This study analyzed the academic performance of three groups of students which are from the years: 2013, 2016, and 2017 and they are undergoing "Additional Educational Obligations" (OFA) since they scored insufficient grades on the admission test. Its objective was to reduce the student's academic failure and its results have been proven to be effective in countering the student's academic ineffectiveness¹⁹.

This study gives support to the purpose of ETugma to provide extra knowledge to the students who are in need of tutoring due to difficulties that are faced during regular class hours.

2.4.2 MAPPING THE TUTORING REFERRAL NETWORK: EXPLORING THE STUDENT-TUTORING CONNECTION

The study identifies the most likely areas that would refer students to or inform students about the tutoring program. Additionally, the study seeks to understand how different students may have interacted with that network of departments in varied patterns of engagement²⁰.

With this study, the proponents can identify the location of where to endorse the system to tutors to further enhance the gathering of data and to identify the needs and specific preferences of students and tutors too.

2.4.3 A Personalized Matching System for Management Teaching Resources Based on Collaborative Filtering Algorithm

This paper develops a personalized matching system for management teaching resources based on collaborative filtering (CF) algorithm. The experimental results show that the proposed CF algorithm can effectively improve the recommendation quality and push personalized teaching resources to each user; the learners are highly satisfied with the personalized matching system²¹.

ETugma's matching module is the main function of the system as it focuses on helping students find tutors and vice versa. Collaborative Filtering helps this module as it refines the recommendations that are made by the system.

2.4.4 Enhancing Recommender Systems: A Hybrid Approach for Precision Matchmaking in Digital Environments

This study is about matchmaking and finding other people that match one's preferences. It uses different algorithms and data analysis such as collaborative filtering and machine learning. It uses these features to contribute to informed decision-making²².

Collaborative Filtering is the algorithm that is used for the recommendations of tutors to students. This algorithm aids students to choose tutors that are the best fit to their preferences in order to create a learning environment that is comfortable for the student to learn in.

2.4.5 Book Recommendation System using Collaborative and Content Based Filtering with Web Scraping

This study discusses using Collaborative Filtering to provide users with suggestions that fits their requirements. It shows that Collaborative Filtering has different uses such as playlist generators for movies or music, product recommendations for online stores, content recommendations for social media²³.

ETugma also uses the concept of finding recommendations based on the user's requirements and preferences. In this way, students can learn in a comfortable environment as their teacher fits the preferences that they have.

2.4.6 Using of CMS Systems for Schools and Education Content Management System for School Information Website

This study proposes the development of content management system (CMS), specifically for school websites. The system aims to simplify the process of

developing and maintaining school websites easier by utilizing the Laravel PHP framework and the Waterfall approach. By providing a CMS platform, schools can efficiently publish and manage information online, enhancing communication with students and the public.

By integrating a CMS in ETugma's system, By organizing content management, we can make it easier for tutors, administrators, and students to update and access important information²⁶.

2.4.7 Implementing Real-Time Notifications in Laravel 11 Using Pusher: A Complete Tutorial

This system which is called Pusher API is used as a way to integrate a chatting system in Laravel as a function of the system.

The chat function is an essential part of ETugma's system as it is the main way of communication of the student to the tutor and also for admins to communicate with the tutors³¹.

METHODOLOGY, RESULTS AND DISCUSSION

3.1 Requirements Analysis

3.1.1 Software Development Life Cycle

The Software Development Life Cycle that will be used for this project is an Agile methodology. Agile methodology is an SDLC that breaks projects into several dynamic phases which are called sprints. In every sprint, the team checks to see if there is anything changed that can help improve their strategy in making the project. There are several phases for an agile methodology which are Plan, Design, Develop, Test, Deploy, Review, and Launch. Planning involves making the requirements needed for the project such as the user stories, objectives, etc. Design is the next phase in which a prototype is made based on the data gathered in the planning phase. Develop is coding the project based on the prototype made. Testing is done to check if the system is functional or if bugs are encountered in its operations. Deploy is a phase in which the project is put in software for other people to use while review is the overall outlook of the project. These phases are done in sprints in which some aspects of the project change to help the team to complete the tasks of the project properly and efficiently. The last phase of the methodology is Launch where the system is completed and published publicly. This methodology is the one chosen by the proponents as the proponents consult with the beneficiary to have a clear understanding of the requirements needed for the system to be better.

AGILE METHODOLOGY



Figure 1. Agile Methodology Diagram

imgsrc: <https://interqualitybg.com/en/resources/scrums-and-agile-resources/agile-methodology>

3.1.2 Conceptual Diagram

INPUT block. It is divided into three parts: Data, Users, and Devices. The Data states the information needed from the users of the system. The Users are the specified users that will use the system. The Devices are the hardwares and softwares that are used to access the system.

PROCESS block. The process block is the development of the researcher's system entitle "ETugma:Tutor Matching System with Assessment for Scribbles and Schemes Early Childhood Center which will help parents to find tutors that will match their preferences.

OUTPUT block. The output blocks display all the output that can be seen by the specific user using the system.

BENEFITS block. This block states the benefits that the users can get from using the system.

The Conceptual Diagram found in Figure 2, shows the IPOB diagram of the research.

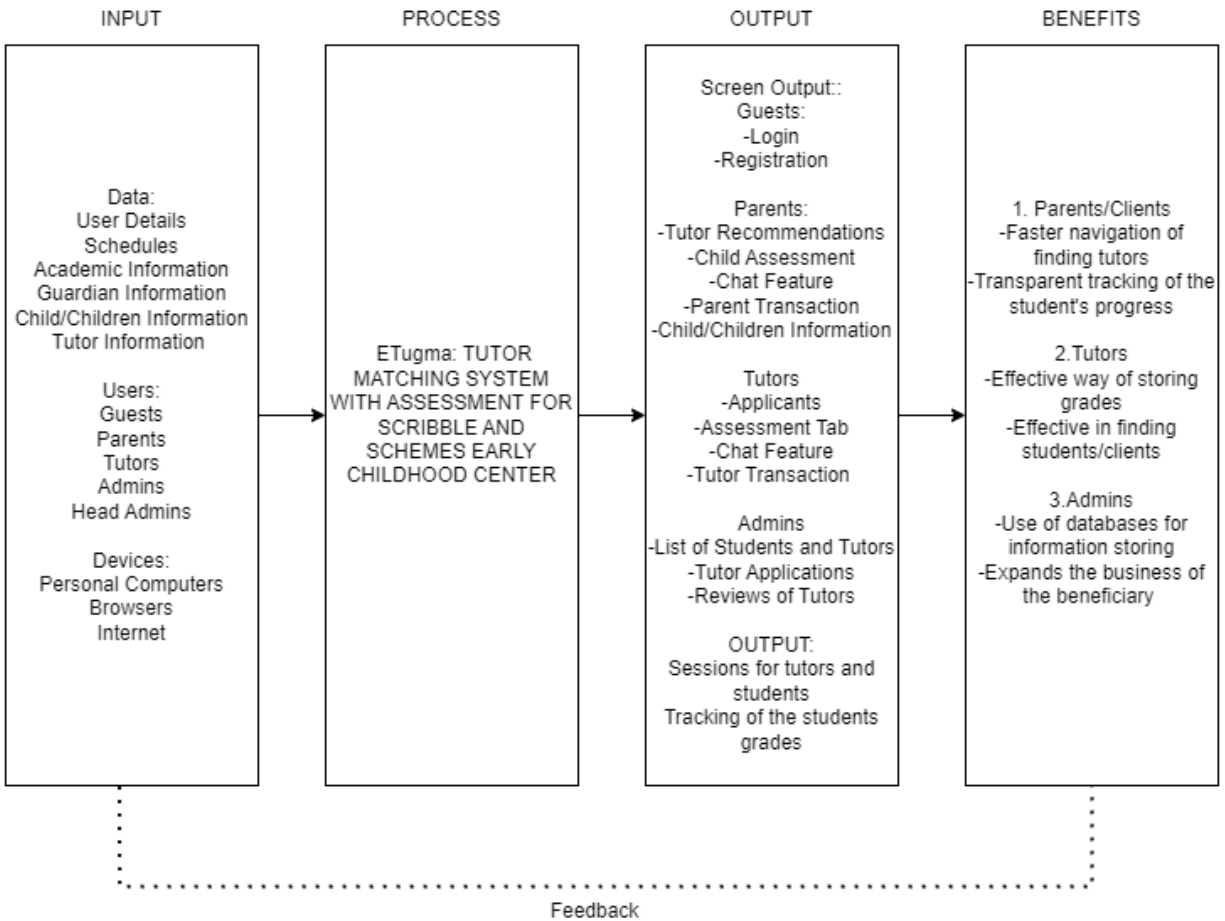


Figure 2. Conceptual Diagram

3.1.3 Flow of Existing System

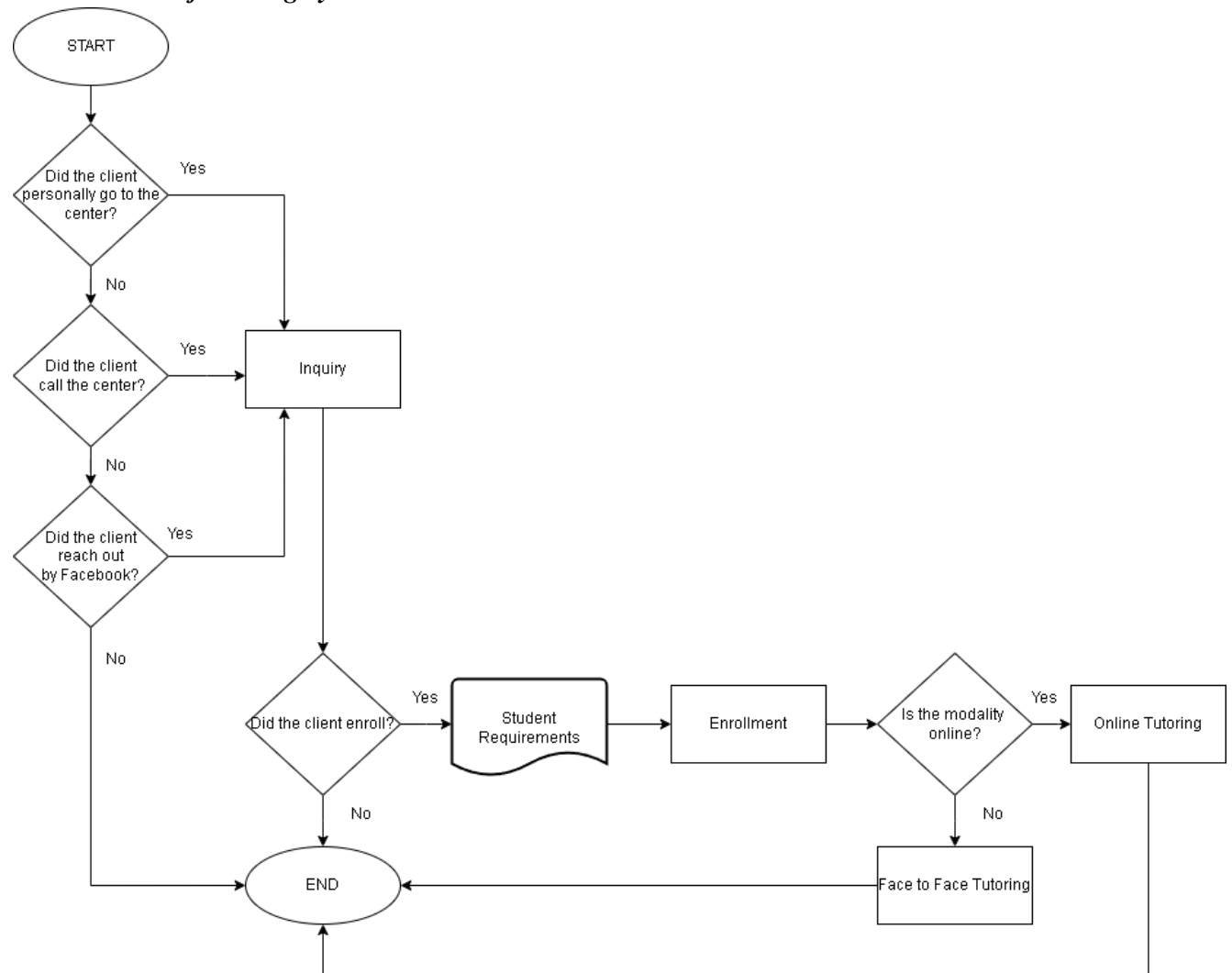


Figure 3. Flow of Existing System

3.1.4 Flow of Proposed System

3.1.4.1 Flowchart of the Guest View

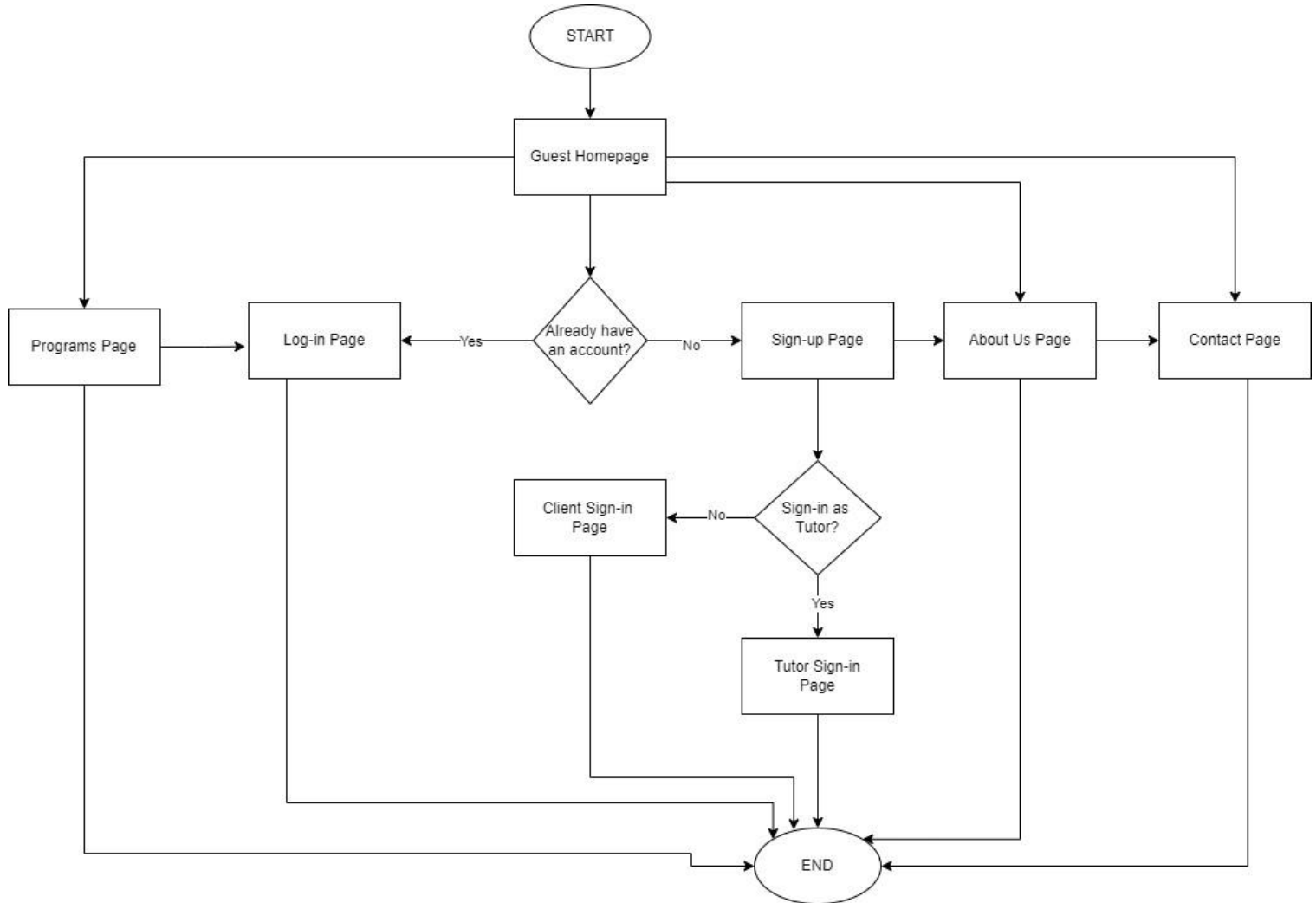


Figure 4. Flowchart of the Guest View

3.1.4.2 Flowchart of the Tutor Registration

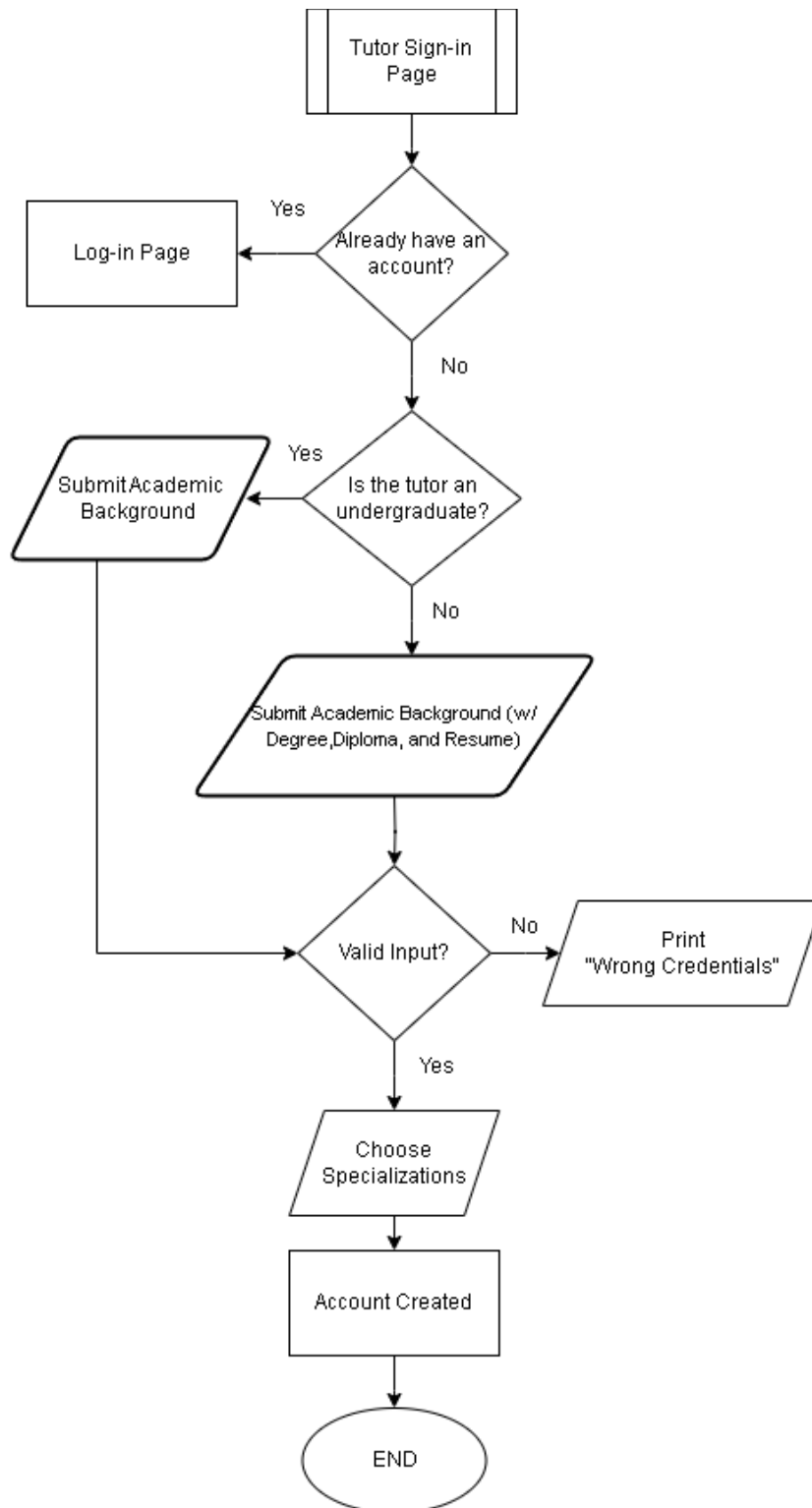


Figure 5. Flowchart of the Tutor Registration

3.1.4.3 Flowchart of the Client Registration

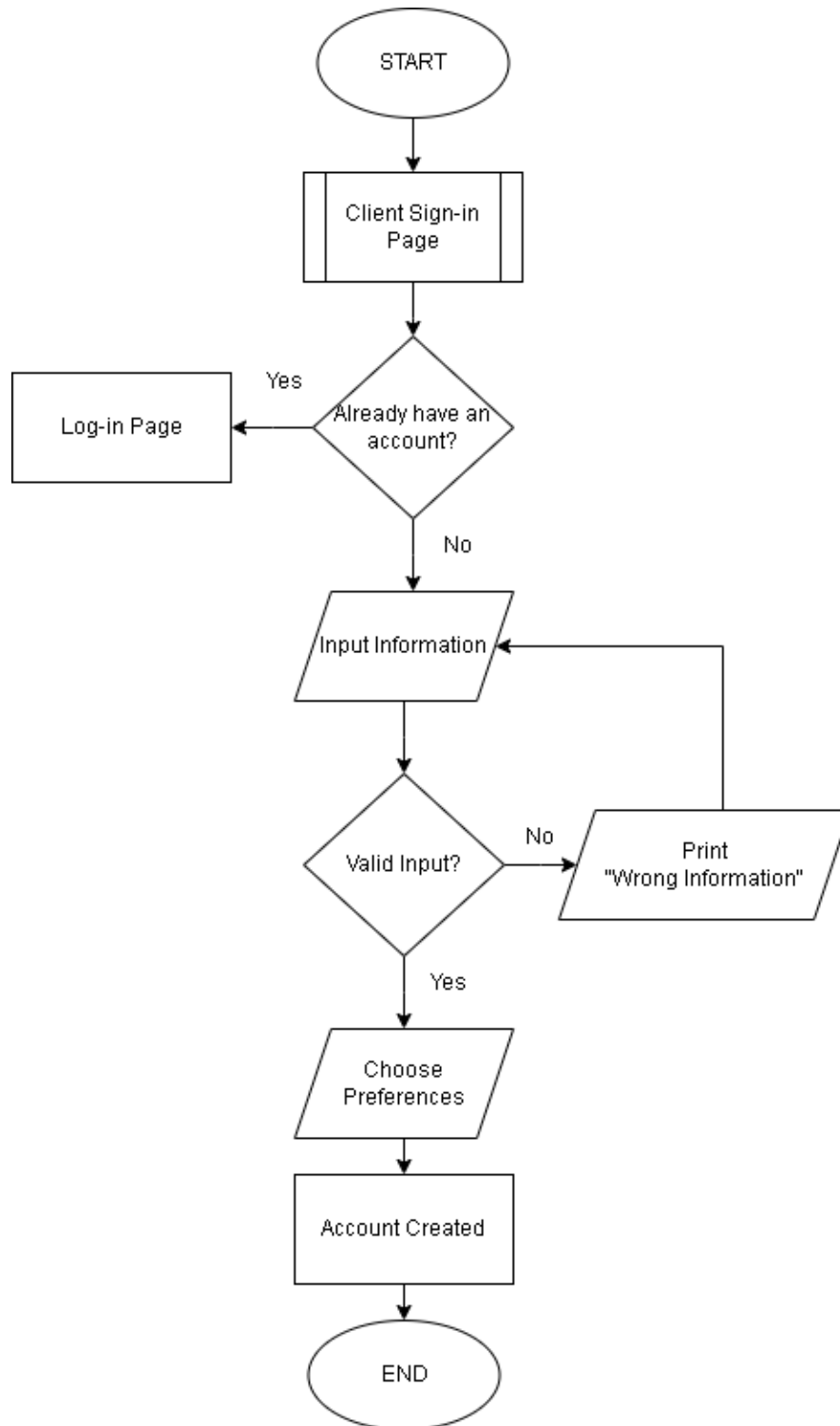


Figure 6. Flowchart of the Client Registration

3.1.4.4 Flowchart of the Client View

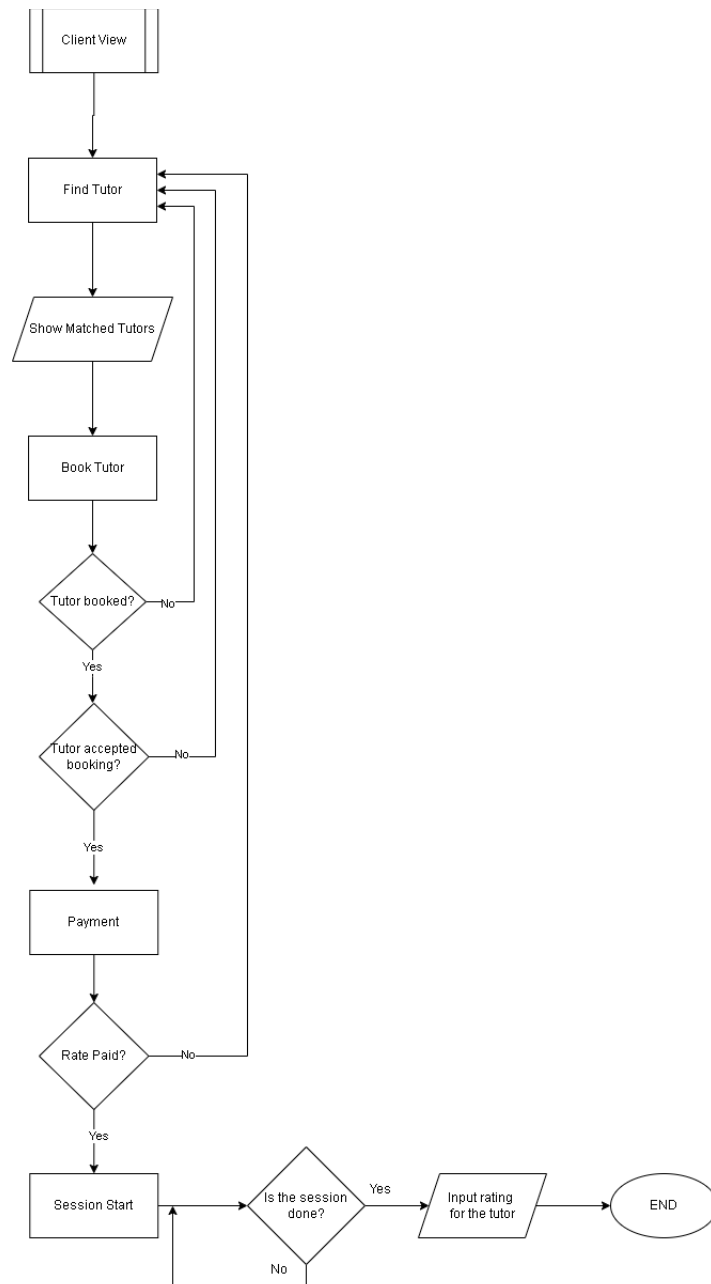


Figure 7. Flowchart of the Client View

3.1.4.5 Flowchart of the Tutor View

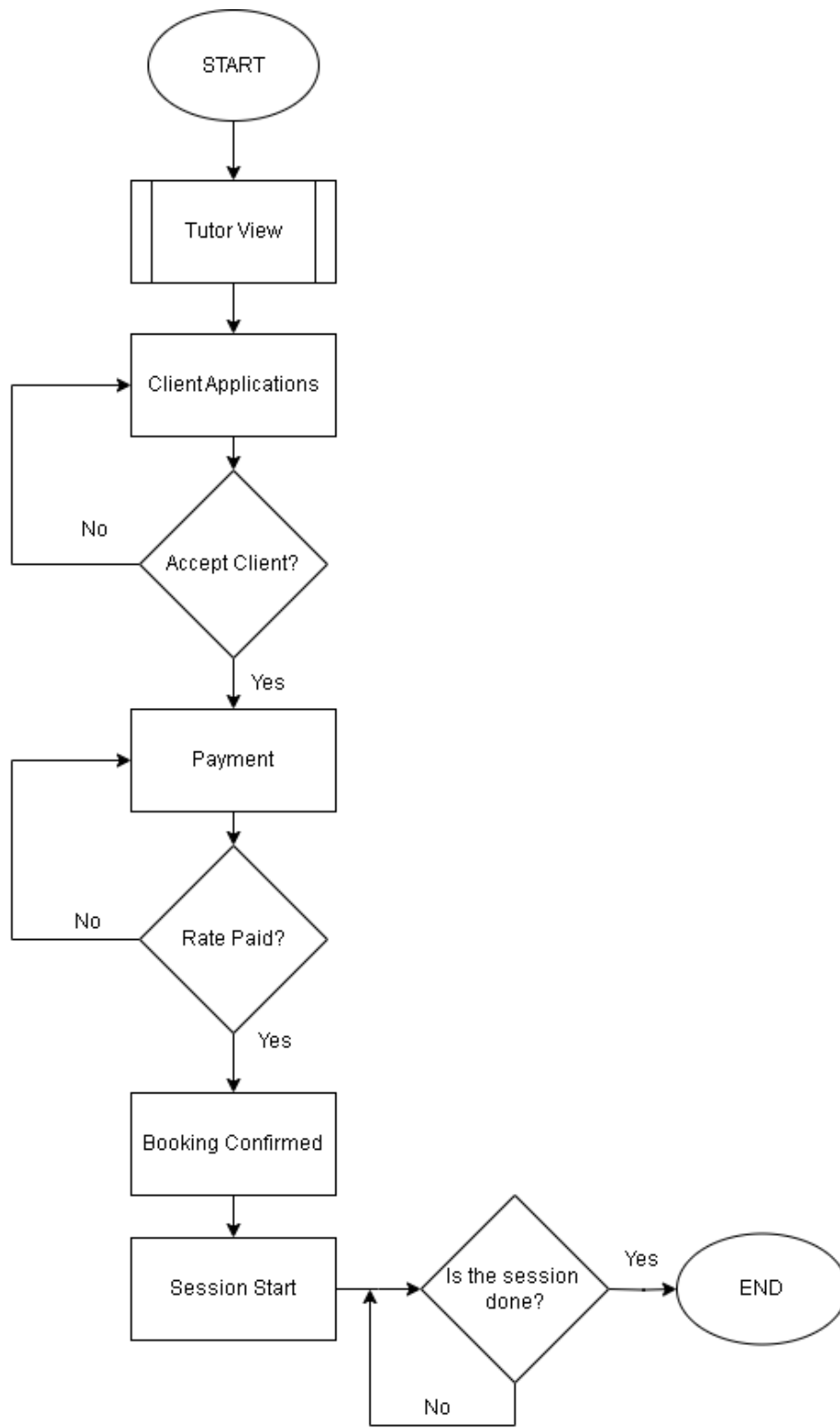


Figure 8. Flowchart of the Tutor View

3.1.4.6 Flowchart of the Admin View

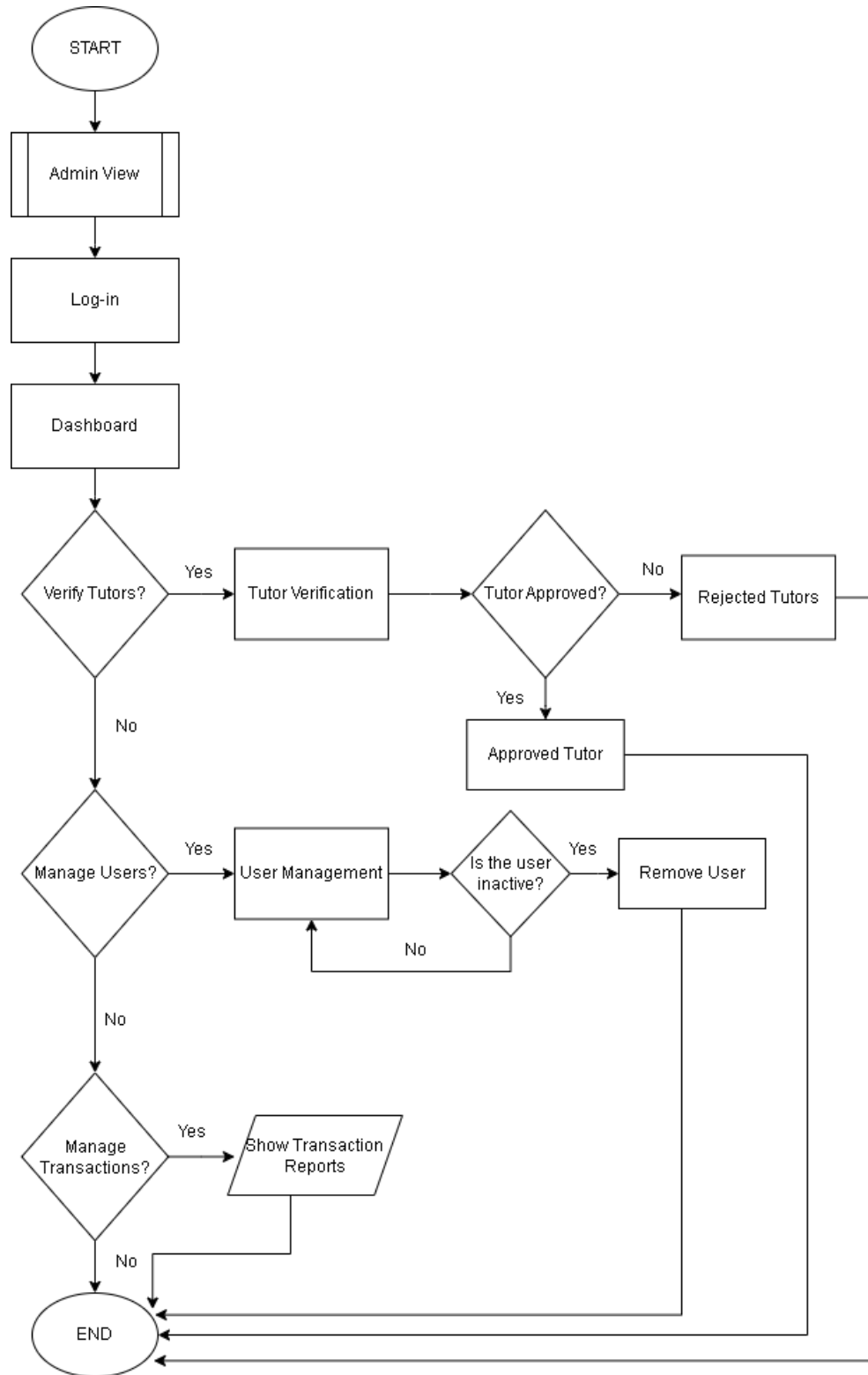


Figure 9. Flowchart of the Admin View

3.1.4.7 Flowchart of the Head Admin View

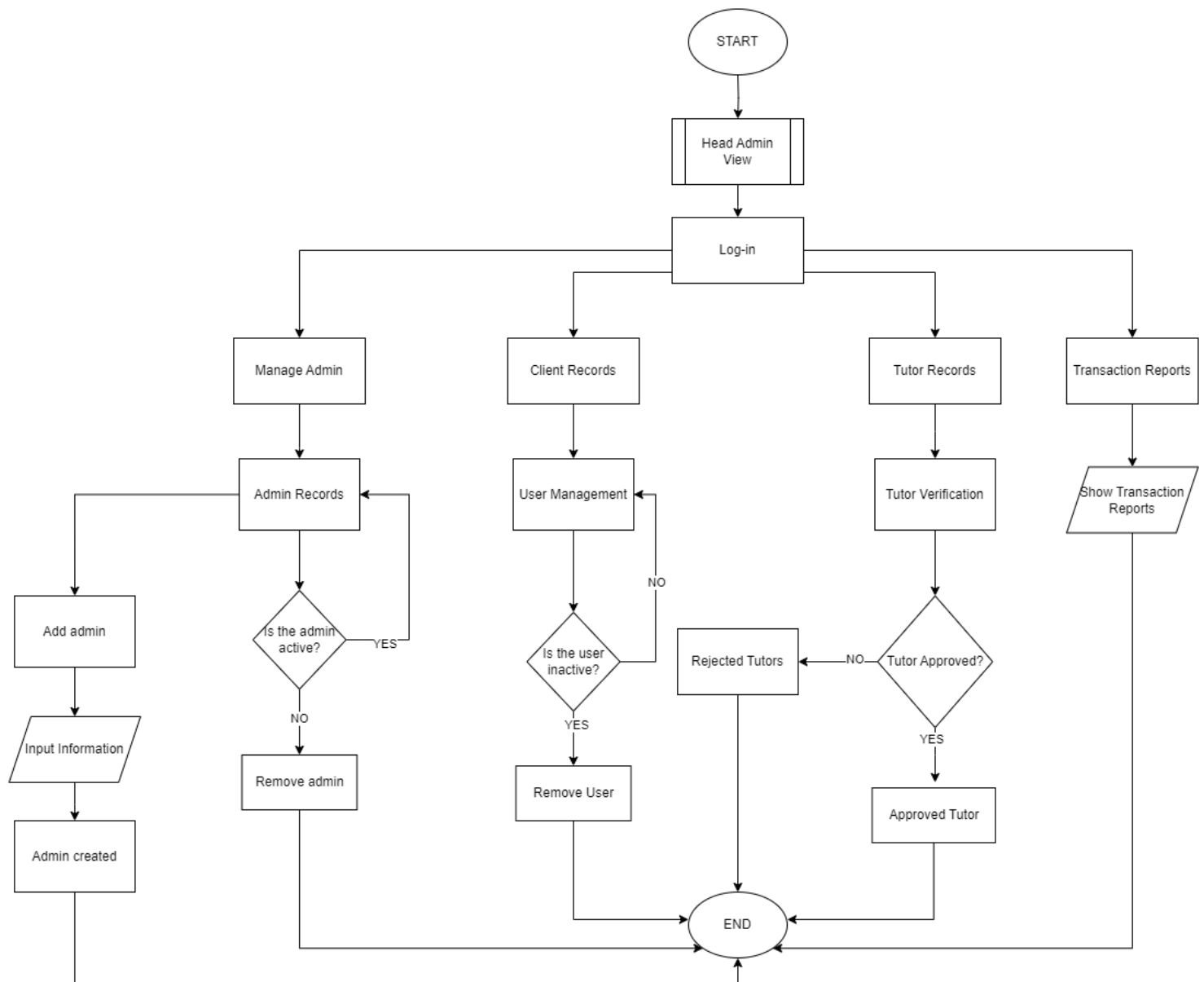


Figure 10. Flowchart of the Head Admin View

3.1.4.8 Data Flow Diagram of the Existing System (Context Diagram)

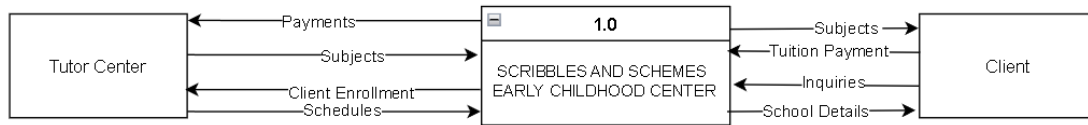


Figure 11. Context Diagram

3.1.4.9 Data Flow Diagram of the Existing System (Level 0)

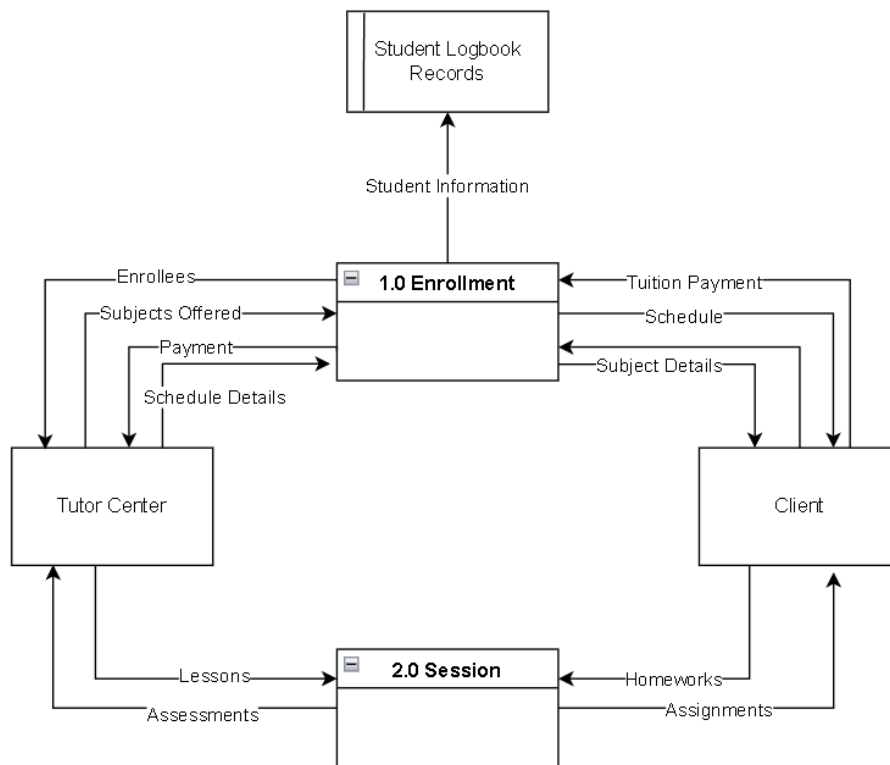


Figure 12. Level 0 Data Flow Diagram of the Existing System

3.1.4.10 Data Flow Diagram of the System (Context Diagram)

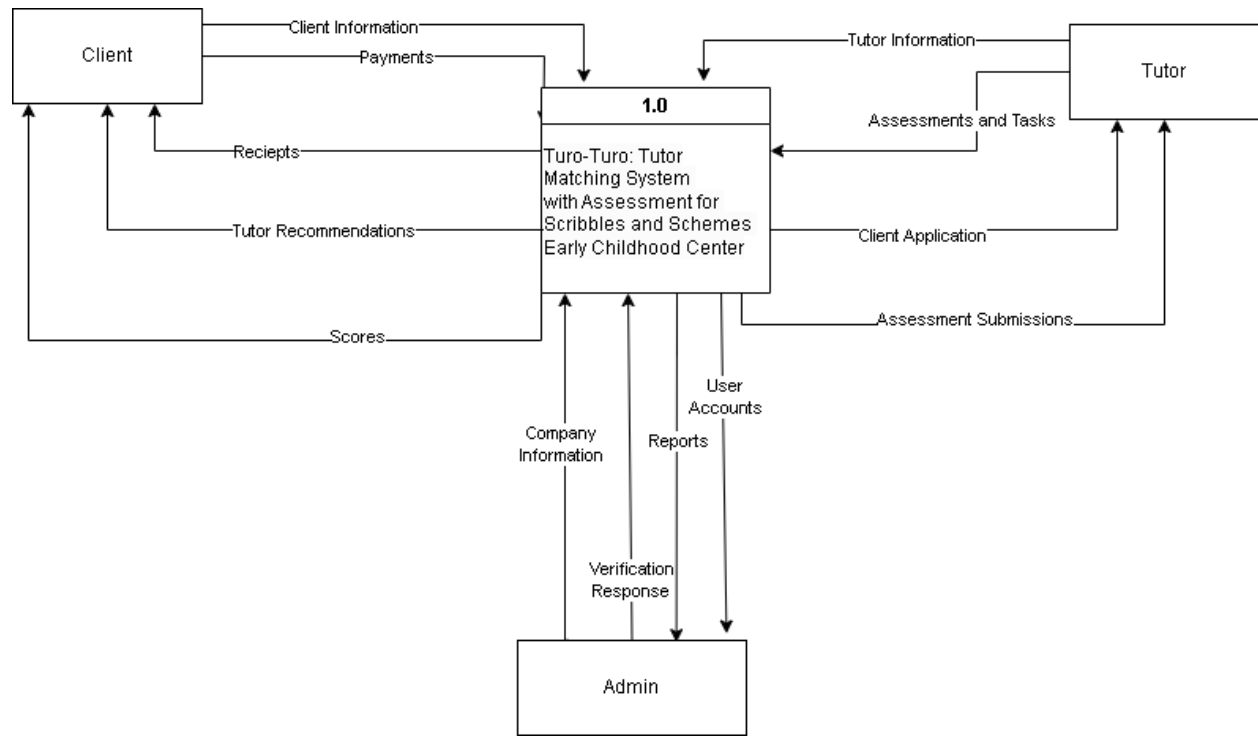


Figure 13. Context Diagram of the Proposed System

3.1.4.11 Data Flow Diagram of the System (Level 0)

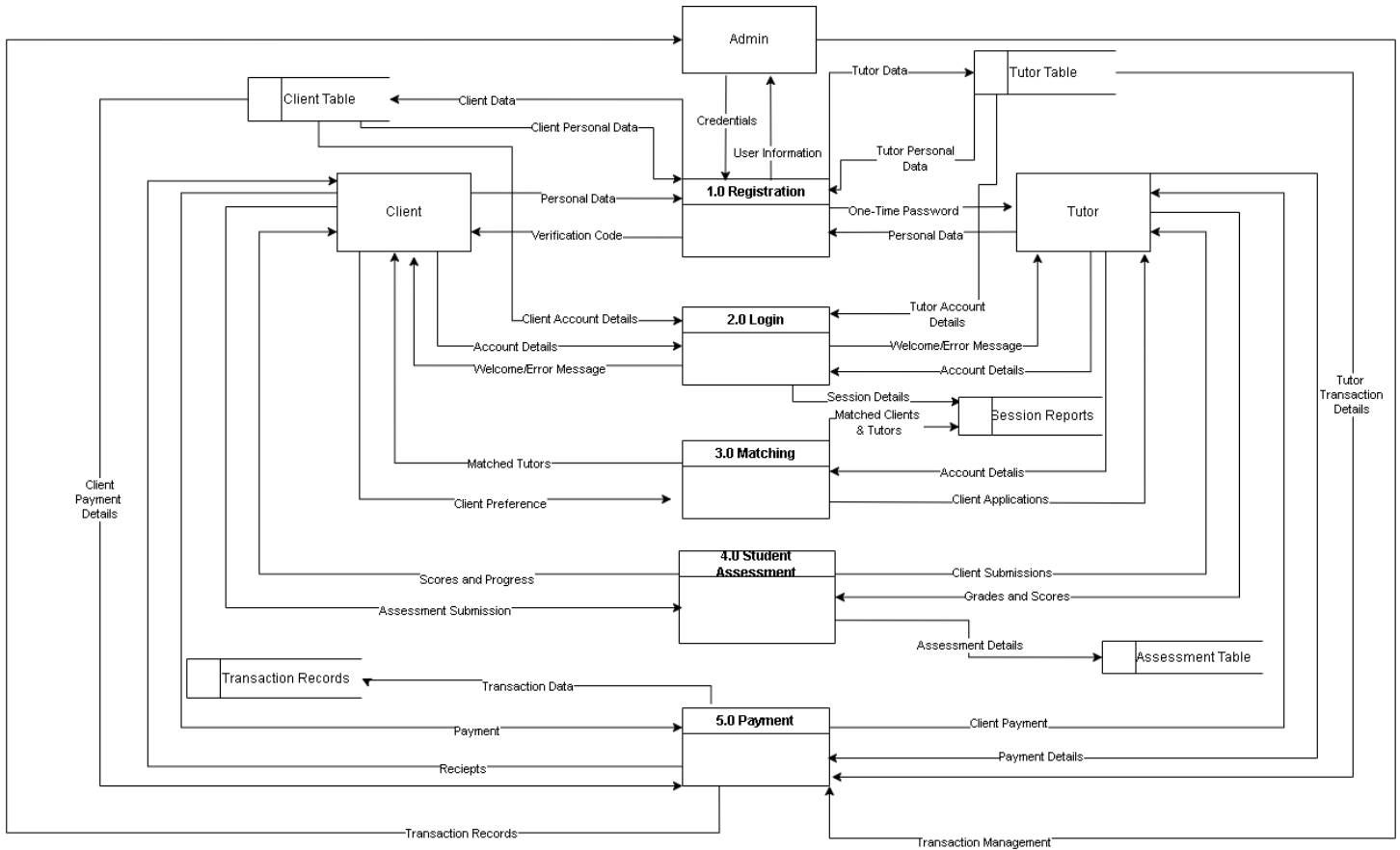


Figure 14. Level 0 Data Flow Diagram of the System

3.2 Requirements Documentation

3.2.1 Methods of Research

In this research, quantitative descriptive research is the main method of research. According to Dovetail, quantitative descriptive research gathers quantifiable data to study the quantifiable data of a population²⁷. These data would show patterns and trends over time, and it is usually seen through polls, surveys, etc.

Using quantitative research can help the researchers in a more detailed way as it gives out structured responses from the population. With the use of surveys to receive ratings from the set of responses given by the researchers, the researchers can gain a conclusion about the responses of the respondents and formulate if the created system is effective and useful for people to use.

3.2.2 Research Instruments

The researchers will use surveys, interviews, and observations as the main research instruments for the system.

Interviews are a research technique in which the researchers will ask open-ended questions to respondents²⁸. In this research, the proponents asked the beneficiary on how their current system works, and the operations of the tutoring center were also observed.

A survey is the collection of information from a sample of individuals using their responses from a questionnaire²⁹. Using surveys, the researchers can formulate an analysis on how the system performs and how the sample users rate it, using the data found from the surveys, it can then be used to formulate a conclusion on whether the system is viable or not.

3.3 Technical Background

Product Description

ETugma is a web-based system that can help parents find a tutor that can cater to the preferences that their children want. It uses a smart algorithm that will make sure that the tutor recommended for the parents matches the preferred characteristics of the student. This system is great for parents who want their children to be tutored with the right teacher.

Economic Use of the Product

ETugma aims to build an effective way for tutors to gain students and vice versa. With this system, students or parents can gain education without having the hassle of scouring through the net for a tutor who can help them with their academic struggles. This system would also help the beneficiary, Schemes and Scribbles Early Childhood Center, expand the reach of their business. This system will integrate the subjects that are being taught by the center and will be centralized on the curriculum they have. With this, the implementation of the ETugma system can benefit the business of the beneficiary.

User Requirements

Head Admin/Admin:

- They must be aware of how to verify the tutor's application.
- They must be knowledgeable on how to monitor each account's actions and to operate the system itself.

Tutors

- They must be a bona fide teacher to be a tutor in the system.
- They must be knowledgeable on how to operate the system.

Students

- They must know how to operate the system.

System Hardware Requirements

OS: Windows 8, 10, 11

RAM: 2GB of RAM; 8GB of RAM recommended.

Storage: At Least 1GB of storage is needed.

System Software Requirements

Internet: 1mbps of internet; 10mbps or more is recommended.

Browsers: Any modern browsers such as Google Chrome, Mozilla Firefox, Opera GX, etc.

3.4 Development and Testing

In the development of the ETugma system, the proponents will use PHP as its main back-end programming language and HTML and CSS as its front-end programming language. Laravel and Bootstrap will be used for the framework of the system. Visual Studio Code is used as the main coding application for the system.

As per testing, it will be tested by ten clients of the Scribble and Schemes Early Childhood Center, our beneficiary. It will then be followed by a survey of the system's functionality and effectiveness using the ISO/IEC 25010.

3.5 Description of the Prototype



Landing Page of the ETugma System

The first page that a user sees when accessing the website.



About Us Page of the System

This page shows what the system does and what the vision and mission of the beneficiary



Programs Offered

Academic Tutorial

Our Center provides a comprehensive Academic Tutorial program tailored for children aged three and above, catering to levels from Pre-School, Elementary, and High School. We offer a variety of subjects, including School Readiness, General Education, and other minor subjects. Our curriculum is designed to foster socio-emotional, physical, language, and cognitive development. By providing a nurturing and engaging learning environment, we aim to build a solid foundation for our students, preparing them for future academic success and personal growth.

[Get started](#) [Subjects & Rates](#)



Partnership and Exclusive Programs

These programs are exclusively available at Scribbles & Schemes - Silang, Cavite branch.



Programs Offered

This page shows the programs that the system offers.



Academic Tutorial Subjects

Pre-School
Elementary
High School

Tutorial Rates

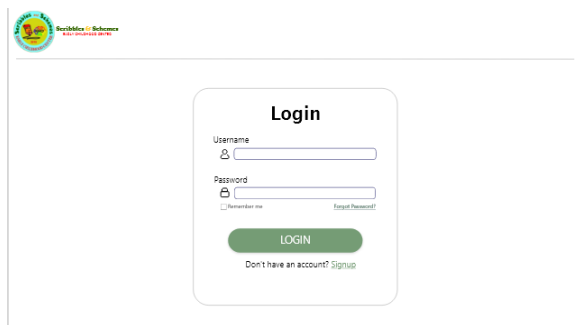
Hourly Sessions	Monthly Sessions
starts at P450	starts at P5,500
Get started	Enroll now

Rates Per Level

	Hourly (Online)	Monthly (Online)	Hourly (F2F)	Monthly (F2F)
Pre-School/Kindergarten				
Grade 1-2				
Grade 3-5				
High School				

Subjects and Rates Page

This page shows the subjects that are offered and what the rates are for those subjects.



Login Page

The Login Page of the system where guests can log in as a client or tutor.



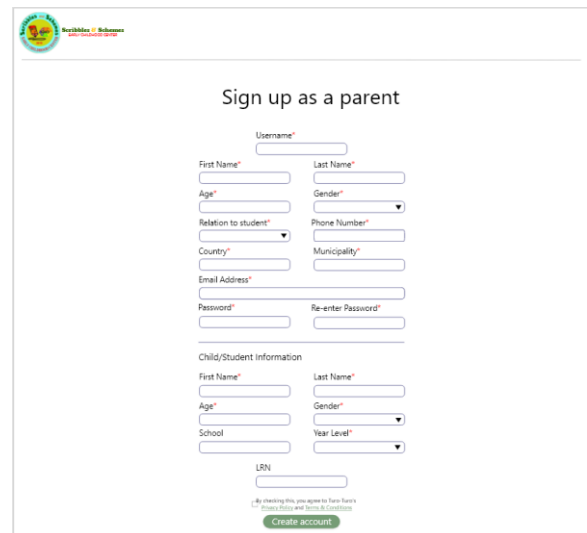
Join as a parent or tutor

[I'm a parent, looking for a tutor](#) [I'm a tutor, looking for a client/student](#)

[Get Started](#)
[Already have an account? Login](#)

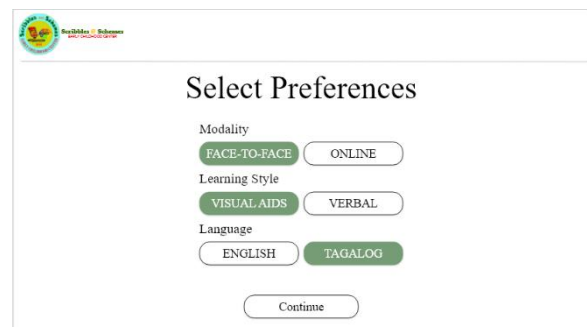
Signup Page

The signup page of the system. Users can sign up as a client or tutor.



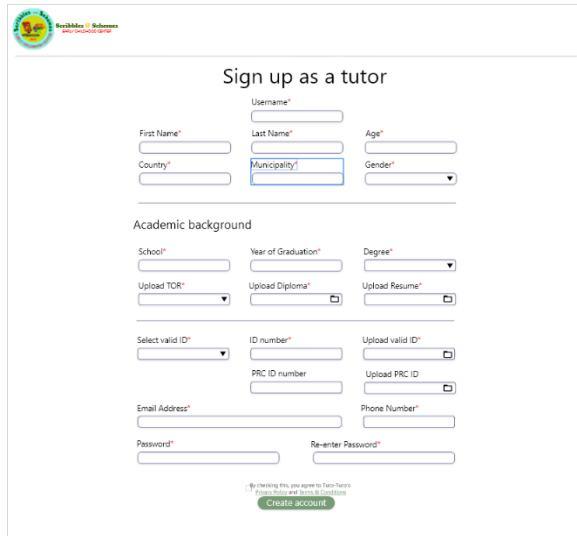
Signup Page for Parents

This page lets the parents register their children for the system.



Select Preferences Page

Clients can select their preference for the tutors they want.



Sign up as a tutor

Username*

First Name* Last Name* Age*

Country* Municipality* Gender*

Academic background

School* Year of Graduation* Degree*

Upload TOR* Upload Diploma* Upload Resume*

Select valid ID* ID number* Upload valid ID*

PRC ID number Upload PRC ID

Email Address* Phone Number*

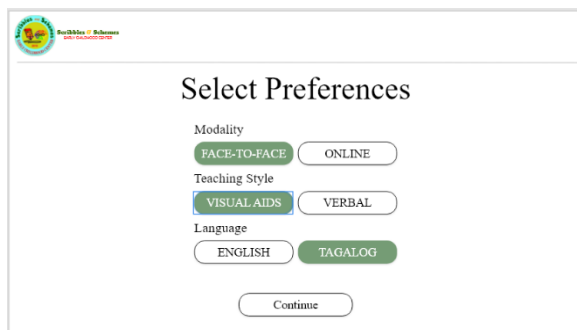
Password* Re-enter Password*

By clicking this you agree to Terms, Privacy Policy and Terms & Conditions

Create account

Sign Up as Tutor Page

This page lets teachers sign up as a tutor for the system. Extra credentials are needed for the verification.



Select Preferences

Modality

FACE-TO-FACE **ONLINE**

Teaching Style

VISUAL AIDS **VERBAL**

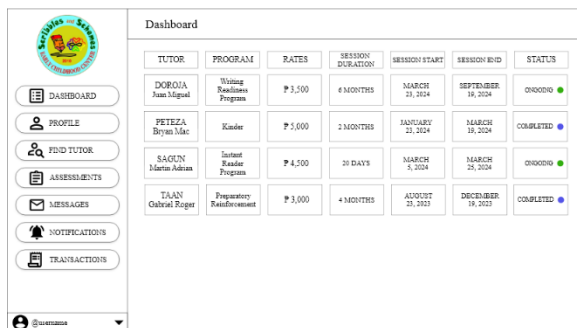
Language

ENGLISH **TAGALOG**

Continue

Select Preferences for Tutors Page

The tutors can select the environment that they are comfortable with while teaching.



Dashboard

TUTOR	PROGRAM	RATES	SESSION DURATION	SESSION START	SESSION END	STATUS
DOROJA Juan Miguel	Writing Readiness Program	P 1,500	4 MONTHS	MARCH 23, 2024	SEPTEMBER 16, 2024	PENDING
PETEZA Bryan Mac	Kinder	P 5,000	2 MONTHS	JANUARY 23, 2024	MARCH 16, 2024	COMPLETED
SAGUN Martin Adrian	Instant Reader Program	P 4,500	30 DAYS	MARCH 5, 2024	MARCH 25, 2024	PENDING
TAAN Gabriel Roger	Preprocessor Reinforcement	P 3,000	4 MONTHS	NOVEMBER 23, 2023	DECEMBER 19, 2023	COMPLETED

Parent/Client Dashboard

This page displays the dashboard of the client. It shows the current status and other information of the sessions they are currently doing.



Juan Miguel Doroja
202400001

Basic Information

Full name Juan Miguel Doroja
Age 12 Years old
Year Level Grade 6
Gender Male

Guardian's Information

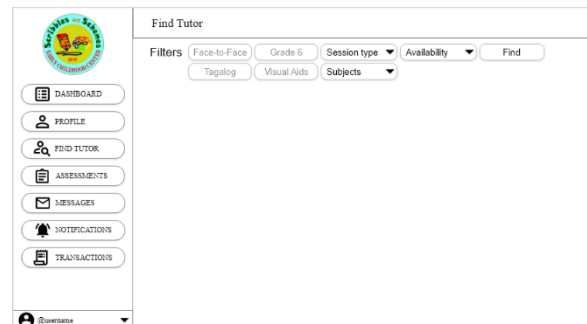
Full name John Mark Doroja
Gender Male
Relation Father
Contact 092723123

STUDENT'S PREFERENCES

Modality: **FACE-TO-FACE** **ONLINE**
Teaching Style: **VISUAL AIDS** **VERBAL**
Language: **ENGLISH** **TAGALOG**

Parent/Client Profile Page

This page displays the information of the client.



Find Tutor

Filters: **Face-to-Face** **Grade 6** **Session type** **Availability** **Find**

Tagalog **Visual Aids** **Subjects**

Find Tutor Page

This page displays the tutors recommended based on the preferences provided by the client.



Deiniel Yanga
Bulacan, Philippines | Tagalog/English | Laguna University

Book Tutor

Credentials

Bachelor's Degree in Information Technology
IT Specialist Database | Cisco Certified

Availability

	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
7-10am	✓	✓	✓	✓	✓	✓	✓
12-1pm	✓	✓	✓	✓	✓	✓	✓
3-5pm	✓	✓	✓	✓	✓	✓	✓



Ratings and Reviews

"He's so good teaching this subject" **2** Open Subjects

"The best teacher who ever taught me this subject" **5** Completed Students

Find Tutor(Tutor's Profile)

This page shows the information of the tutor that can be found in the find tutor page.

Book a Tutor

Deiniel Yanga

Session Type: ☐ Face-to-Face ☐ Online

Choose Date and Time: (date picker) (time picker)

Year level:

Subject:

Topic:

If Face-to-Face, preferred location:

Book Tutor

Book a Tutor Page

After checking the tutor's information, clients can book the tutor.

Messages

Search User:

LERI MEDINA

View Messages

Message Tab for Client

This page displays the messages of the client.

Notifications

Tutor: Tutor sent you a chat.

System Notification: Booking request status.

System Notification: Session ended, please assess your son/daughter.

System Notification: Booking request status.

Son: Son sent you a chat.

System Notification: Booking request status.

View tutor

Notifications Tab for Client

This page shows the notifications of the clients.

STUDENTS	PROGRAM	RATES	SESSION DURATION	SESSION START	SESSION END	STATUS
DOROJA Juan Miguel	Writing Fundamentals Program	P 3,500	8 MONTHS	MARCH 23, 2024	SEPTEMBER 18, 2024	PENDING
PETEZA Bryan Mac	Kindergarten	P 5,000	2 MONTHS	JANUARY 23, 2024	MARCH 18, 2024	COMPLETED
SAGUN Martin Adrian	Instant English Program	P 4,500	20 DAYS	MARCH 5, 2024	MARCH 25, 2024	PENDING
TAN Gabriel Roger	Preparatory Readiness Program	P 3,000	4 MONTHS	AUGUST 23, 2023	DECEMBER 18, 2023	COMPLETED

Dashboard for Tutors

This dashboard shows the dashboard of the tutor. This shows the information of their current sessions.

Juan Miguel Doroja

ABOUT ME: lorem ipsum lorem ipsum

SUBJECT OFFERED: Mathematics

TUTORING STYLE: ☐ Visual aids ☐ Verbal

MODALITY: ☐ Face-to-Face ☐ Online

ACHIEVEMENTS: Bachelor's Degree in Information Technology, IT Specialist Database, CISCO Certified

AVAILABILITY:

	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
7-10am	✓	✓	✓	✓	✓	✓	✓
12-1pm	✓	✓	✓	✓	✓	✓	✓
3-5pm	✓	✓	✓	✓	✓	✓	✓

TUTORING PRICE:

Subject	Price
Math	Php150
Science	Php150
GMRC	Php150

Tutor Profile Page

This page shows the profile of the tutors. It displays the information that they have.

CLIENT APPLICATIONS

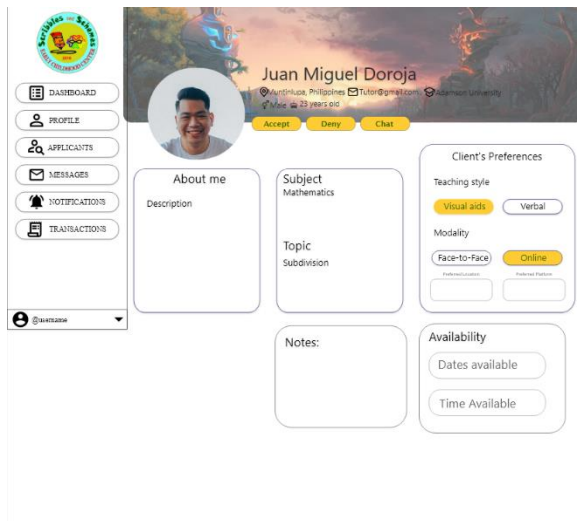
Juan Miguel Doroja

John Michael Doroja

View more

Client Applications Page

This page shows the clients that are applying to book the tutor.



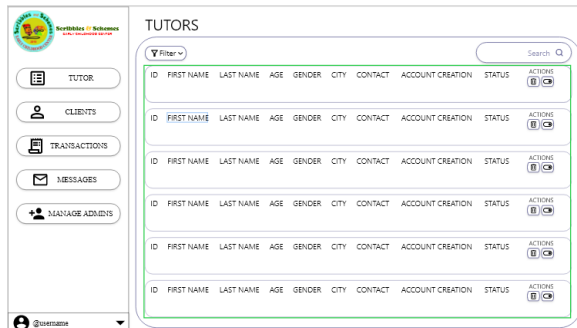
Client Application (View More)

This page shows extra information about the client.

The screenshot displays a user profile for Juan Miguel Doraja. The profile includes a profile picture, name, and contact information. Below the profile, there are sections for 'About me', 'Subject' (Mathematics), 'Topic' (Subdivision), 'Notes', 'Client's Preferences' (Teaching style: Visual aids, Verbal; Modality: Face-to-Face, Online), and 'Availability' (Dates available, Time Available).

Client Application (View More)

This page shows extra information about the client.



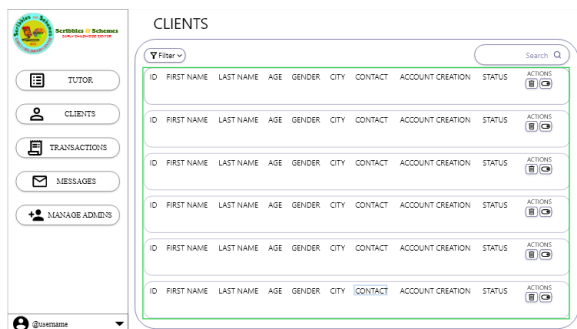
Head Admin - Tutors Tab

This tab can show the list of tutors registered in the system. They can also manage the accounts from this tab.

The screenshot shows a table of tutors with columns: ID, FIRST NAME, LAST NAME, AGE, GENDER, CITY, CONTACT, ACCOUNT CREATION, STATUS, and ACTIONS. The table is filtered by 'TUTOR' and shows a list of tutors with their respective details and action buttons.

Head Admin - Tutors Tab

This tab can show the list of tutors registered in the system. They can also manage the accounts from this tab.



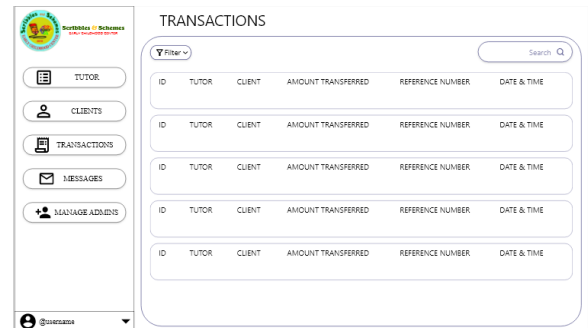
Head Admin - Client Tab

This tab can show the list of client registered in the system. They can also manage the accounts from this tab.

The screenshot shows a table of clients with columns: ID, FIRST NAME, LAST NAME, AGE, GENDER, CITY, CONTACT, ACCOUNT CREATION, STATUS, and ACTIONS. The table is filtered by 'CLIENTS' and shows a list of clients with their respective details and action buttons.

Head Admin - Client Tab

This tab can show the list of client registered in the system. They can also manage the accounts from this tab.



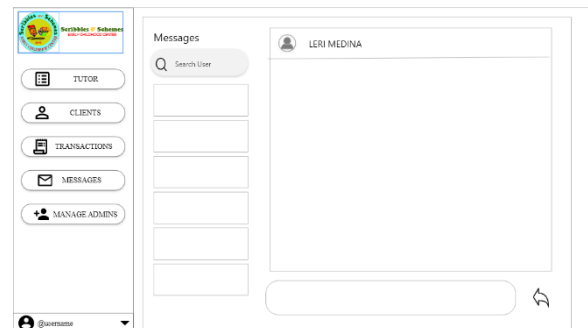
Head Admin-(Transaction Tab)

This tab shows the transactions that have happened between the client and tutors.

The screenshot shows a table of transactions with columns: ID, TUTOR, CLIENT, AMOUNT TRANSFERRED, REFERENCE NUMBER, and DATE & TIME. The table is filtered by 'TRANSACTIONS' and shows a list of transactions with their respective details.

Head Admin-(Transaction Tab)

This tab shows the transactions that have happened between the client and tutors.



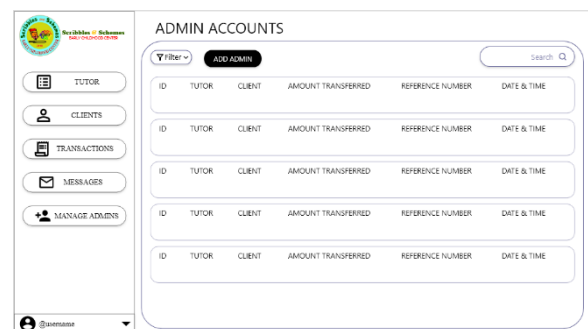
Head Admin-(Messages Tab)

This tab shows the messages the head admin has done.

The screenshot shows a message interface with a search bar and a list of messages. The messages are filtered by 'MESSAGES' and show a list of messages with their respective details.

Head Admin-(Messages Tab)

This tab shows the messages the head admin has done.



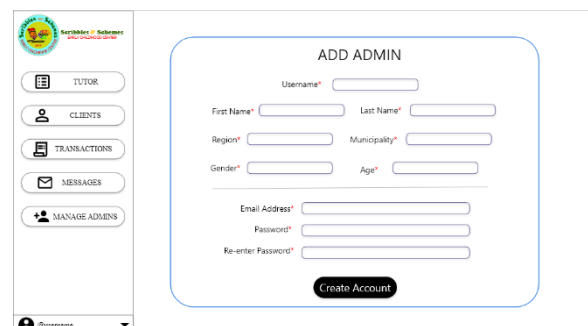
Head Admin-(Admin Accounts)

This tab can show the list of admins registered in the system. They can also manage the accounts from this tab.

The screenshot shows a table of admin accounts with columns: ID, TUTOR, CLIENT, AMOUNT TRANSFERRED, REFERENCE NUMBER, and DATE & TIME. The table is filtered by 'ADMIN ACCOUNTS' and shows a list of admin accounts with their respective details.

Head Admin-(Admin Accounts)

This tab can show the list of admins registered in the system. They can also manage the accounts from this tab.



ADD ADMIN

This form is used to add a new admin. It includes fields for Username, First Name, Last Name, Region, Municipality, Gender, Age, Email Address, Password, and Re-enter Password. A 'Create Account' button is at the bottom.

Head Admin - (Add Admin Tab)

This tab lets the head admins add new admins in the system.

Admin - Tutors Tab

This tab can show the list of tutors registered in the system. They can also manage the accounts from this tab.

Admin - Clients Tab

This tab can show the list of clients registered in the system. They can also manage the accounts from this tab.

Admin - Verification Tab

This tab can verify whether the applying tutor can teach at the system or not.

Admin-(Messages Tab)

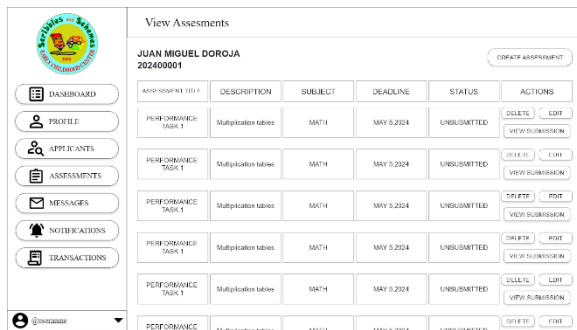
This tab shows the messages the admin has done.

Admin-(Transaction Tab)

This tab shows the transactions that have happened between the client and tutors.

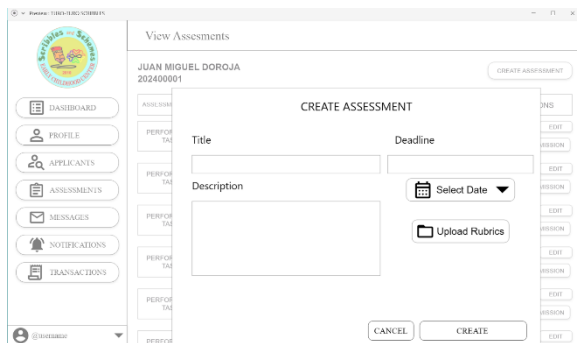
Tutor Assessment Tab

This tab lets the tutor to either view or create an assessment for the student.



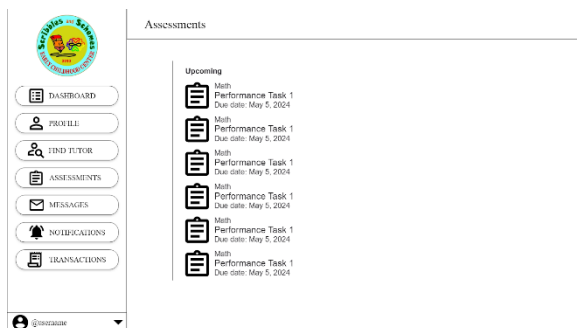
Tutor- View Assessment Tab

In this tab, tutors can view the status of the assessments that have been made.



Tutor-Create Assessment

This tab allows the tutor to create an assessment for their student.



Client-Assessment Tab

This tab lets the client see their current assessments.

3.6 Implementation Plan

The ETugma system aims to build a bridge for both tutors and students to create an effective learning environment. With the matching algorithm of the system, clients can look up the tutors that they are confident of helping them with their academic struggles while tutors can use their strengths as a teacher to guide their students in their studies. This

system will also help tutors in their transactions as it provides them ways to keep track of their students grades and the assessments that they require of the students. All of these records will be kept using a database for safekeeping and fast retrieval of the data needed.

For the smooth operation of the system, the administrators of the system will be given training on how to operate and maintain the system. The staff of Scribbles and Schemes Early Childhood Center will be the first administrators of the system as they are the beneficiary of the project.

In the maintenance of the system, the developers will regularly check and visit the beneficiary to see if updates will need to be done.

3.7 Implementation Results

For the implementation results, the proponents will expect that the system will serve its objectives such as providing a learning environment for both tutor and client.

4. SUMMARY, CONCLUSION AND RECOMMENDATIONS

4.1 Summary

Evaluation of the proposed system using ISO 25010 Quality Model

Indicators	Mean	Interpretation
A. Functionality		
Suitability		
B. Performance		
Efficiency		
C.		
Compatibility		
D. Usability		
E. Reliability		
F. Security		
G.		
Maintainability		
H. Portability		
Total Mean		

During the initial testing of the ETugma system, the system score “” based on the ISO 25010 Quality Model evaluation. These scores are because of such:

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